

Siddh Mandirwala

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Professional Experience

Software Engineer Intern, AI4Purpose Inc. [🔗](#) | New York, NY Feb 2026 – Present

- Spearheaded migration from WordPress to Next.js with TailwindCSS, achieving 60% reduction in page load times and improving maintainability.
- Architected 10+ responsive, mobile-first components using Next.js App Router, reducing cross-device layout bugs by 80% and achieving WCAG 2.1 compliance.
- Delivered 15+ frontend features across 6 Agile sprints with Git/GitHub, increasing deployment frequency by 3x through streamlined code reviews.

Software Development Engineer Intern, KeyToZ | Surat, India Jan 2024 – Jul 2024

- Developed healthcare platform features using Vue.js, React.js, and Node.js/Strapi, driving end-to-end testing via Playwright in CI/CD pipelines.
- Debugged and resolved 12+ critical production issues across frontend and backend, improving mobile responsiveness by 40%.
- Optimized database queries and RESTful API endpoints, reducing server response times by 25% and streamlining data retrieval.

Full Stack Developer Intern, Bharat Tech Labs | Surat, India Jul 2023 – Aug 2023

- Built social media scheduling platform using Vue.js, Node.js, Redis, PostgreSQL, and Express.js with automated publishing workflows.
- Designed Redis-based job queue processing 500+ scheduled posts daily with 99.5% uptime via distributed task architecture.
- Established error handling and logging using Docker-containerized microservices, reducing production incidents by 30%.

Education

Master of Science - MS, Computer Science, New York University Sep 2024 – May 2026

- Current Grade - 3.815 / 4.0
- Courses - Algorithms, AI, ML, Data Science, Big Data, Software Engineering, Information Security & Privacy, Visualization
- Student Engagement Ambassador - Raised \$12,000+ support for scholarships.

Bachelor of Engineering - BE, Computer Engineering, Gujarat Technological University Sep 2020 – Jun 2024

- Grade - 9.20 / 10.0 (3.68 / 4.0)
- Courses - Data Structures & Algorithms, OOP, Cloud Computing, Computer Network, Web Development

Skills

Programming Languages: Python | TypeScript/JavaScript | Java | C/C++ | SQL

Frontend: Next.js | React.js | Vue.js | Tailwind CSS | HTML5 | CSS3 | D3.js | Framer Motion

Backend & APIs: Node.js | Express.js | Django | RESTful APIs | GraphQL | Supabase | Strapi

Databases: PostgreSQL | MySQL | MongoDB | Redis | Qdrant

ML & AI: NLP | RAG | Predictive Analytics | scikit-learn | PyTorch | XGBoost | LLM | Vercel AI SDK

DevOps, Cloud & Data: Git/GitHub | Docker | Kubernetes | AWS | Vercel | CI/CD | Apache Spark | PySpark | Pandas

Projects

Portafina [🔗](#),

- Engineered responsive portfolio with Next.js 16, React 19, TypeScript, and Framer Motion featuring dark/light theme support.
- Integrated AI chatbot powered by Google Gemma 3 4B via Vercel AI SDK with real-time streaming and anti-hallucination prompting.
- Set up lead capture with Supabase, enforcing client/server-side rate limiting (3 queries/guest, 6/user) to prevent API misuse, and optimized SEO.

RAGStack [🔗](#),

- Created end-to-end RAG system aggregating data from GitHub, Medium, and LinkedIn with two-stage text normalization pipeline.
- Fine-tuned GPT-2 using PEFT over 50 epochs on Kaggle with Weights & Biases tracking, published weights to Hugging Face.
- Orchestrated microservices via Docker Compose with MongoDB, Qdrant for vector retrieval, Flask/FastAPI, and Gradio interface.

Vizpromax [🔗](#),

- Created interactive visualization platform analyzing decade-long NYPD complaint data across NYC boroughs with precinct-level analysis.
- Designed 8 narrative-driven D3.js visualizations exploring spatio-temporal crime trends, deployed on Vercel for urban planning insights.

MTAnalytics [🔗](#),

- Built distributed analytics pipeline using Apache Spark and PySpark processing 100GB+ of NYC MTA data across 472+ stations over 17 years.
- Developed station-level ridership forecasting using XGBoost with SHAP interpretability, visualized with Plotly, Matplotlib, and Folium.